

Problem

For an FM radio receiver, the incoming wave is in the frequency range from 88 to 108 MHz. The tuner circuit is a parallel RLC circuit with a $4\text{-}\mu\text{H}$ coil. Calculate the range of the variable capacitor necessary cover the entire band.

Step-by-step solution

Step 1 of 3

Given that, $L = 4\text{-}\mu\text{H}$

The frequency range for FM radio receiver is 88MHz to 108 MHz. we consider the low and high ends of band.

Resonant frequency $(\omega_0) = \frac{1}{\sqrt{LC}}$

$$\omega_0^2 = \frac{1}{LC}$$

$$(2\pi f_0)^2 = \frac{1}{LC}$$

$$4\pi^2 f_0^2 = \frac{1}{LC}$$

$$\therefore C = \frac{1}{4\pi^2 f_0^2 \times L}$$

Step 2 of 3

For the high end of the FM radio receiver,

$f_0 = 108\text{ MHz}$ and the corresponding 'C' is

$$C_1 = \frac{1}{4\pi^2 \times (108 \times 10^6)^2 \times 4 \times 10^{-6}}$$

$$\therefore C_1 = 0.543\text{ PF}$$

Step 3 of 3

For the low end of the FM radio receiver,

$f_0 = 88\text{ MHz}$, and the corresponding 'C' is,

$$C_2 = \frac{1}{4\pi^2 \times (88 \times 10^6)^2 \times 4 \times 10^{-6}}$$

$$\therefore C_2 = 0.818\text{ PF}$$

Thus, C must be adjustable capacitor varying from **0.543 PF** to **0.818 PF**.